

Research paper

A hybrid chaotic bat algorithm for optimal placement and sizing of dg units in radial distribution networks

Imene Khenissi^a, Tawfik Guesmi^{b,*}, Badr M. Alshammari^b, Khalid Alqunun^b, Abdulaziz Almalaq^b, Mansoor Alturki^b, Rafik Neji^a

^a Department of Electrical Engineering, LETI Laboratory, National Engineering School of Sfax, University of Sfax, Tunisia

^b Department of Electrical Engineering, College of Engineering, University of Ha'il, Ha'il 55211, Saudi Arabia

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ABSTRACT

Appropriate location and sizing of distributed generation (DG) units in a radial distribution system (RDS) play a significant role in the improvement of its overall performance. Indeed, proper DG placement and sizing help in maintaining the balance between power supply and demand, decreasing energy losses and enhancing voltage profile. Within this context, this paper presents an effective approach for the determination of the optimal location and the appropriate capacity of a photovoltaic distributed generation (PVDG) unit in RDSs. In the suggested approach, the best position of the PVDG is selected using the loss sensitivity factor (LSF). Meanwhile, a new optimization technique incorporating chaos, bats' self-adaptive compensation, and Doppler effect into the original bat algorithm (BA) is developed to find the optimal size of the PVDG unit. The optimal PVDG size is optimally determined in such a way that total active power losses of the RDS is reduced and voltage profile is enhanced. The performance of the proposed optimization technique, symbolized by (CSA-DC)BA, is evaluated using various benchmark functions. Moreover, the applicability and effectiveness of the suggested approach are verified on the IEEE 33-bus and the IEEE 69-bus RDSs. Obtained results revealed that the proposed (CSA-DC)BA outperforms the comparison optimization techniques.

1. Introduction

Due to the world population growth, providing reliable and secure electricity demand with high quality and low cost is considered one of the most electrical utilities challenges. In fact, increasing load demand leads to a power loss increase and voltage drop in the RDS (Alzaidi et al., 2019). These two issues are the main causes of power quality decrease of RDSs. In order to address these issues, several methods and techniques have been applied in the literature such as, capacitors integration to compensate the reactive power (Astrid et al., 2022; Abdelaziz et al., 2016), network reconfiguration (Sellami et al., 2022; Kandasamy et al., 2022; Vai et al., 2021), distributed generation (DG) installation (Mohammadi and Esmaili 2012; Khenissi, Sellami et al., 2021; Khenissi, Fakhfakh et al., 2021; Kashyap et al., 2022) and so on.

Since they are connected near the loads at strategic points to enhance the electrical system reliability and to expedite customer's power delivery, DG systems acquire a crucial potential in power distribution systems. Based on the power that can provide, DG systems can be

classified into three categories: (1) only real power injected DG systems, (2) only reactive power injected DG systems and (3) active and reactive injected DG system (Essallah et al., 2018). For instance, DG units based on PV, fuel cells and micro-turbine are belonging to the first category. DG units based on synchronous compensators are examples of the second category. Regarding the third category, which has the capability to inject both active and reactive powers, includes DGs equipped with synchronous machines. Generally, the installation of DGs is widely used and applied for power losses issue. It is considered as the most accurate solution to mitigate power losses increase (Alzaidi et al., 2019), since these sources generate clean and reliable power supply and deliver reactive and active powers (Essallah et al., 2018). However, in accordance with other research works, inappropriate integration of DG units can affect the frequency stability of the power system and cause voltage fluctuations and instability in the distribution network. In addition, if DG units are improperly placed and sized, they may cause a reverse power flow into the grid, which can lead to technical challenges, such as increase of system power losses and voltage deviation (Khenissi et al.,

* Corresponding author.

E-mail address: tawfik.guesmi@istmt.rnu.tn (T. Guesmi).

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2023). Accordingly, proper placement and allocation of DG units are essential to mitigate these issues and ensure a reliable and efficient distributed generation system deviation (Khenissi et al., 2023).

1.1. Literature review

Choosing the optimal location and finding the optimal capacity of DG units in distribution systems have attracted the interest of several researchers. Indeed, a wide range of methods and approaches have been developed for solving the DG placement and sizing problem. Generally, these approaches include analytical methods (Kumar et al., 2018; Wu et al., 2022), numerical methods (Gallego Pareja et al., 2022; Abou El-Ela et al., 2021), optimization-based methods (Truong et al., 2022; Moradi and Abedini, 2012) and hybrid techniques (Jagan Mohan and Arul, 2017; Sheng et al., 2015). Analytical techniques, generally based on sensitivity analysis, have demonstrated their effectiveness in dealing with several optimization problems related to power systems such as, power flow analysis (Chen et al., 2016), network reconfiguration (Li et al., 2019; Abdelkader et al., 2021), voltage control (Alzaareer et al., 2022) and optimal placement of FACTS devices (Zand et al., 2023). Generally, such methodologies involve reducing losses within the distribution system and enhancing voltage profiles. For instance, an improved analytical technique based on a high-convergence optimization method has been investigated in (Kumar et al., 2018) to efficiently allocate and size a DG system in the IEEE 7-bus and IEEE 71-bus RDSs. The aim of this technique was to reduce system power losses and improve its voltage profile. Loss sensitivity factor (LSF) has been introduced in (Wu et al., 2022) to select the best placement of PV and wind turbine (WT) systems combined with an energy storage system in the IEEE 33-bus distribution network. The primary focus of this method was to minimize grid power losses. In (Azad et al., 2021), an innovative analytical algorithm has been presented to determine the most suitable placement and capacity of DGs in RDSs. To do this, a novel combined index (CI) focused on minimizing active power losses and enhancing the system's voltage profiles has been developed to find the suitable location of the DGs. Then, a search algorithm has been developed to find the optimal sizing of the DGs. The proposed methodology has been evaluated through the IEEE 12-bus and 33-bus RDSs.

Numerical methods have been also proposed in several research studies for investigating such problems (Gallego Pareja et al., 2022; Abou El-Ela et al., 2021). For instance, in (Gallego Pareja et al., 2022), a mixed-integer linear programming (MILP) based method has been proposed for an optimal reconfiguration of a distribution network and optimal DG placement. In this method, power flow problem has been linearized and then solved by CPLEX software. According to that reference, the application of linearization is driven by the need to address the limitations inherent in nonlinear methods when dealing with larger and more complex RDSs. While simulation results showed its robustness, the proposed MILP based method comes with certain limitations and shortcomings such as loss of precision and accuracy in the obtained solutions, especially for large systems. Other numerical method based on the dynamic programming has been introduced in (Abou El-Ela et al., 2021) to place and determine appropriate capacities of DG units, with the objective of reducing network losses and enhancing the voltage profile.

In recent years, the utilization of meta-heuristic techniques for solving complex optimization problems has significantly increased due to their efficiency in handling high-dimensional search spaces. These techniques, inspired by natural processes, have shown remarkable capabilities in discovering near-optimal solutions for large and non-linear real-world problems where classical methods are insufficient (Truong et al., 2022; Moradi and Abedini, 2012). In this sense, various meta-heuristic techniques were developed and improved for handling the optimal sizing and placement of DG units in RDSs. For illustration, (Moradi and Abedini, 2012) have proposed a hybrid approach that combines genetic algorithms (GA) with particle swarm optimization

(PSO) to determine the best placement and capacity of DG in RDSs. The primary aim was to reduce overall power losses, enhance voltage regulation, and improve voltage stability where various operational and security constraints have been incorporated in the problem formulation. This multi-objective optimization problem has been converted into mono-objective optimization problem by using a weighted-sum approach. Other combined PSO-GA algorithm has been proposed in (Jagan Mohan and Arul, 2017) for the same purpose. In (Sheng et al., 2015), the optimal allocation and sizing of multiple DG units have been achieved using an improved version of the non-dominated sorting GA-II (NSGA-II). The suggested multi-objective optimization method has been applied to simultaneously minimize the total line losses, minimize voltage deviation and maximize voltage stability margin. In this method, mutation and crossover operators in the differential evolution algorithm have been used instead of the original NSGA-II operators to increase its convergence rate and improve the quality of the optimal non-dominated solutions. In the same context, it has been noted that an increasing amount of research studies has been focused on investigating and introducing more recent ameliorated and developed optimization methods to properly allocate DGs systems such as, the coot optimization algorithm (COOA) (Kien et al., 2022), the Archimedes optimization algorithm (AOA) (Elseify et al., 2023), the modified firefly algorithm (MFA) (Bhadoriya and Gupta, 2021), the transient search optimization algorithm (TSOA) (Sukraj et al., 2019), the modified flower pollination (MFP) algorithm (Ramshanker et al., 2022), and the modified crow search algorithm (Grisales-Noreña et al., 2023), which have shown and demonstrated their effectiveness when dealing with this kind of problem.

Recently, bat algorithm (BA) has received considerable attention due to its effectiveness in addressing intricate multi-objective optimization issues across diverse domains, including power systems (Yuvaraj et al., 2017). BA is a nature-inspired optimization technique proposed by Xin-She Yang in 2010 and it mimics the echolocation behavior of bats in finding prey and avoiding obstacles. However, like other meta-heuristic techniques, the standard BA is considered as an efficient method in the exploitation phase, while its exploration capacity is considered weak. In fact, this algorithm suffers from its weak performance in terms of convergence to the global optimum solution, since it locally drops to the optimal solution (Liu et al., 2020). Hence, notwithstanding its advantages, the convergence performance of the BA should be further ameliorated and improved. To do this, various variants of BA have been proposed in the literature to enhance its efficiency, convergence speed, and effectiveness in solving complex optimization problems (Dash et al., 2022). Recently, the amelioration of the BA has been occurred based on two aspects (Dash et al., 2022). The hybridized BA (HBA) based methods where BA is mixed with other concepts and features adopted from other algorithms (Ravikumar et al., 2019; Bahmani-Firouzi and Azizipannah-Abarghoee, 2014) present the first aspect, while the second aspect includes the modified BA (MBA) techniques (Reddy et al., 2022; Azad et al., 2021). Due to the high time complexity of the most HBA based methods (Dash et al., 2022), the MBA techniques have been widely used and introduced by researchers (Rachid et al., 2021). In various MBA methods, self-adaptation mechanisms have been incorporated to dynamically adjust their parameters during the optimization process. The self-adaptive approaches aim to eliminate the need for manual parameter tuning and make the algorithm more robust and effective across different optimization problems. It allows the algorithm to autonomously refine its behavior by ensuring a balance between exploration and exploitation (Kouicem et al., 2022) which leads to an improved optimization performance. In (Reddy et al., 2022), a novel self-adaptive learning method has been proposed for the probabilistic application of four velocity updating strategies within BA. The main aim of this method was to improve the movement and exploration of bats in the search space. This improved BA was applied for optimal sizing of a hybrid energy conversion system incorporating both PV and wind turbine (WT) systems along with battery energy storage. However, only the

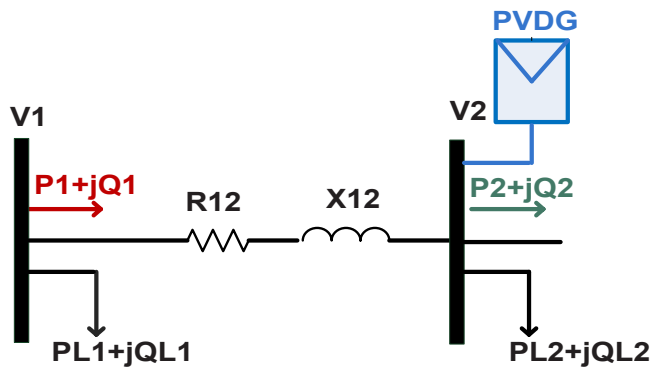


Fig. 1. A distribution line in RDS.

total operating cost was considered and no considerable attention has been given to the power losses reduction. In (Meng et al., 2015; Saxena et al., 2021), an improved BA has been proposed to further imitate the bats' behaviors. To do this, the habitat selection of bats as well as their self-adaptive compensation for Doppler effect in echolocation have been integrated into the original BA. By introducing the habitat selection, bats can look for their foods in diverse foraging habitats compared to the basic algorithm. The habitat selection has been modeled as the choice between quantum and mechanical behaviors. Besides, by integrating the self-adaptive compensation, bats can adaptively compensate depending on their targets, for the Doppler Effect in echoes. Hence, simulation results have proved that the improved self-adaptive compensation for Doppler effect based BA outperformed the standard BA algorithm and other optimization methods in terms of solutions diversity and convergence characteristics. This technique will be symbolized in this study by (SA-DC)BA. However, like other stochastic techniques, (SA-DC)BA uses random variables in its phases which can affect the optimal solution quality.

In recent years, the incorporation of chaos theory into the optimization algorithms has demonstrated a remarkable advancement in the stochastic optimization field (Nguyen et al., 2021; Zhu and Zheng 2019). One of the fundamental features of chaos systems is their extreme sensitivity to initial conditions, known as the "butterfly effect" where small changes in these conditions create large changes in the results. Classic stochastic optimization methods, like PSO, GA and BA often use standard probability distributions, such as uniform and Gaussian distributions to generate random numbers. However, chaos can offer an alternative approach to introduce randomness and aid in exploration due to its ergodicity and mixing properties. Indeed, it has been shown that chaos can contribute to faster and potentially more effective iterative search steps compared to traditional stochastic search methods using standard probability distributions (Nguyen et al., 2021). Chaos can be incorporated into optimization techniques in many ways. For example, in reference (Nguyen et al., 2021), a chaotic local search has been added to the original symbiotic organisms search algorithm to help prevent convergence to local optima and improve the exploration of the search space. The proposed optimization method has been applied to find the optimal reconfiguration of RDS. In (Zhu and Zheng, 2019), a chaotic perturbation of the current solutions of PSO algorithm has been applied to improve the obtained solution quality. This can introduce randomness and avoid the convergence to the local optima, by favoring a large exploration of the search space.

1.2. Main contributions

As far as the authors are aware, limited efforts have been made to combine the sensitivity analysis with meta-heuristic optimization algorithms for finding the optimal position and appropriate capacity of the DG systems. Furthermore, the introduction of BA and its derivatives to

solve this issue has not been covered in depth. Within this context, the main key contributions of this research paper compared to other methods investigated and illustrated in the aforementioned references can be summarized as follows.

- A new methodology to optimally size the PVDG system and to properly locate it in an RDS is suggested. In the suggested methodology, the optimal location and sizing of the PVDG are determined by simultaneously minimizing the system real power losses and voltage deviation. Due to the nonlinear objective functions, discrete and continuous decision variables, the studied problem is formulated as a multi-objective mixed-integer nonlinear programming (MOMINP). This MOMINP is solved in two phases. Firstly, the candidate node for the PVDG system is selected using a loss sensitivity based method. Then, a novel meta-heuristic optimization technique hybridizing the (SA-DC)BA with chaos theory is implemented to determine the optimal size of the PVDG system corresponding to minimum system real power losses and voltage deviation. It is to be noted that minimization of the voltage deviation aims to improve the voltage profile by shifting voltage magnitudes of all RDS buses as close as possible to one per unit (pu). In the proposed optimization technique, symbolized by (CSA-DC)BA, various chaotic maps are used to generate chaotic sequences to replace the random numbers involved in the generation of the initial solutions, the quantum behavior, the mechanical behavior and the local search processes of the (SA-DC)BA algorithm.

Since the optimal placement and sizing of PVDG is handled as a multi-objective minimization problem, there is a set of optimal non-dominated solutions instead of unique solution. Thus, a fuzzy-based method is employed to extract the appropriate compromise solution among the obtained solutions.

- A comparison study between the proposed (CSA-DC)BA technique and other optimization algorithms such as, PSO, GA, ABC, (SA-DC)BA, HPSO (Aman et al., 2014), MOPSO (Aman et al., 2013), DGPI (Memarzadeh and Keynia, 2020) and CBOM (Memarzadeh et al., 2023) is illustrated to demonstrate the superiority of this new optimization method.

- To further validate and test the applicability and robustness of the suggested methodology, two test systems which are the IEEE 33-bus and the IEEE 69-bus RDSs are adopted.

1.3. Paper organization

The reminder of this paper is organized as follows. The problem formulation is introduced in Section 2. Section 3 illustrates the loss sensitivity analysis for the optimal PVDG allocation. The proposed optimization method is presented in Section 4. Section 5 exhibits the numerical results using several benchmark functions. Simulation results are developed and discussed in Section 6. The conclusion is presented in Section 7.

2. Problem formulation

The main aim of this study is to determine the optimal placement and size of a PVDG system in an RDS with the objective of decreasing the overall system losses and minimizing voltage deviation (VD). To reach these targets, a special attention to the objective function formulation is required with a specific limitations satisfaction. As mentioned earlier, due to the high R/X ratio in RDSs, conventional power flow methods are considered inappropriate techniques for identifying load flow solutions. Consequently, the forward-backward sweep method is used in this paper owing to its robustness, simplicity, speed and low memory requirement (Yavaraji et al., 2021). Let's consider a distribution line (1,2) connecting two buses "1" and "2", as presented in Fig. 1, where bus "1" is chosen to be the sending end bus and bus "2" is the receiving end bus. P_1 and Q_1 represent the active and reactive power flow through the branch (1,2), respectively. PL_1 and QL_1 represent the active and the reactive powers consumed by the load connected to bus 1. R_{12} and X_{12} represent the

resistance and the reactance of the branch (1,2), respectively. V_1 and V_2 are the voltages at buses 1 and 2, respectively.

The optimal allocation and sizing of PVDG unit in the RDS is converted into a multi-objective optimization problem where several equality and inequality constraints are simultaneously considered.

2.1. Objective functions

Two objective functions, which are real power loss minimization and VD improvement, are considered in the studied optimization problem.

2.1.1. Line losses minimization

Before DGs integration in the RDS, the system power flow equations can be described by Eqs. (1) and (2) (Astrid et al., 2022).

$$P_1 = P_2 + PL_1 + R_{12} \times \left(\frac{P_1^2 + Q_1^2}{|V_1|^2} \right) \quad (1)$$

$$Q_1 = Q_2 + QL_1 + X_{12} \times \left(\frac{P_1^2 + Q_1^2}{|V_1|^2} \right) \quad (2)$$

After the integration of the DG unit in the RDS, the previous power flow equations in the branch (1,2) will be as given by expressions (3) and (4) (Mahmoud et al., 2020).

$$P_1 = P_2 + PL_1 + \left(R_{12} \left(\frac{P_1^2 + Q_1^2}{|V_1|^2} \right) \right) - P_{DG} \quad (3)$$

$$Q_1 = Q_2 + QL_1 + \left(X_{12} \left(\frac{P_1^2 + Q_1^2}{|V_1|^2} \right) \right) - Q_{DG} \quad (4)$$

The real and reactive power losses in the branch (1–2) can be calculated using Eq. (5).

$$P_{loss}(1,2) = R_{12} \times \left(\frac{P_1^2 + Q_1^2}{|V_1|^2} \right) \quad (5)$$

The total real power losses (TP_{loss}) in the RDS can be determined by summing the real power losses across all branches. Thus,

$$TP_{loss} = \sum_{j=1}^{N-1} P_{loss}(j,j+1) \quad (6)$$

where N is the number of the system buses $P_{loss}(j,j+1)$ is the real power losses through the branch ($j,j+1$).

2.1.2. Voltage profile amelioration

It is noteworthy to highlight that the voltage profile is affected by the intermittency of the connected load. Moreover, with the penetration of DG systems, the voltage quality and stability in the distribution system are considered unpredictable. Hence, inappropriate allocation and capacity of DGs can cause voltage rise and dips at all system buses. Within this context, the second objective function is used to ameliorate the voltage profile. Indeed, an acceptable voltage profile achieves a reliable and stable power delivery to the consumers. Improvement of voltage profile can be achieved by minimizing VD expressed by the following equation (Kumar et al., 2022).

$$VD = \min \sum_{i=1}^N |1 - V_i|^2 \quad (7)$$

where V_i presents the voltage magnitude at the i th node.

2.1.3. Weighted-sum approach

In order to implement the proposed optimization algorithm, the studied multi-objective optimization problem is transformed into a single objective optimization problem using the weighted-sum

approach. To do this, both objective functions are normalized to ensure that they are with a similar scale. Then, different weights are assigned to these functions to evaluate their importance during the optimization process.

$$F = \omega_1 TP_{loss} + \omega_2 \rho VD \quad (8)$$

where

$$\omega_1 + \omega_2 = 1 \quad (9)$$

and

$$\rho = \frac{TP_{loss}^{max}}{VD^{max}} \quad (10)$$

ω_1 and ω_2 are the weighting factors for the total real power losses and voltage deviation functions, respectively. Coefficient ρ is used to make both objectives with similar scale. It is the ratio of the maximum value of the total real power losses (TP_{loss}^{max}) to the maximum voltage deviation ($\frac{TP_{loss}^{max}}{VD^{max}}$).

Since the optimal placement and sizing of PVDG unit in RDS is treated as a multi-objective minimization problem, there is no unique solution that optimally minimizes all objectives simultaneously. Indeed, it is required to find a set of optimal non-dominated solutions representing the trade-offs between the objective functions to be minimized (Almalaq et al., 2023). These solutions also called Pareto solutions can be obtained by varying ω_1 and ω_2 from 0 to 1. It is worth noting that, the minimum real power losses is obtained when $\omega_1=1$ and $\omega_2=0$, while, the minimum voltage deviation is obtained when $\omega_1=0$ and $\omega_2=1$. In order to facilitate the decision making process for selecting the appropriate compromise solution among the obtained Pareto solutions, a fuzzy-based method is used to describe the decision maker's satisfaction (Almalaq et al., 2023). Due to the imprecision and uncertainty of the makers' decision, the satisfactory level of a Pareto solution X_i for the j th objective function (F_j) is described by the following membership function.

$$\mu_j^i = \begin{cases} 1, & \text{if } F_j(X_i) \leq F_j^{\min} \\ \frac{F_j^{\max} - F_j(X_i)}{F_j^{\max} - F_j^{\min}}, & \text{if } F_j^{\min} < F_j(X_i) < F_j^{\max} \\ 0, & \text{if } F_j(X_i) \geq F_j^{\max} \end{cases} \quad (11)$$

where $F_j^{\min}$ and $F_j^{\max}$ are minimum and maximum values of F_j , respectively.

The normalized membership function evaluated at a solution X_i is expressed as follows.

$$\mu^i = \frac{\sum_{j=1}^m \mu_j^i}{\sum_{i=1}^s \sum_{j=1}^m \mu_j^i} \quad (12)$$

where m is the number of the objective functions and s is the number of Pareto solutions. The best compromise solution is the non-dominated solution having the maximum value of μ^i .

2.2. Problem constraints

2.2.1. Power balance constraints

The power balance equations describe the real and reactive power flow equations, which can be expressed by the following equality constraints.

$$P_R + \sum_{i=1}^a P_{PVDG}(i) = \sum_{j=1}^N PL_j + \sum_{k=1}^c P_{loss}(k) \quad (13)$$

$$Q_R + \sum_{i=1}^a Q_{PVDG}(i) = \sum_{j=1}^N QL_j + \sum_{k=1}^c Q_{loss}(k) \quad (14)$$

where P_R and Q_R represent the active and reactive powers injected from the slack bus, respectively. a and c are the total number of PVDG system and system branches, respectively.

2.2.2. PVDG position constraint

The optimal allocation of PVDG system on the radial distribution network should satisfy the following equation (Kumar et al., 2022).

$$2 \leq PVDG_{placement} \leq N \quad (15)$$

In fact, the first bus is considered as the slack or substation bus. Subsequently, the optimal placement of the PVDG system should not be taking into consideration at this bus.

2.2.3. Voltage profile constraint

The voltage intermittency at each node should lay in maximum and minimum limits as given in Inequality (16) (Kumar et al., 2022).

$$V^{\min} \leq V_i \leq V^{\max} \quad (16)$$

Typical values of $V^{\min}$ and $V^{\max}$ are 0.9 p.u. and 1.1 p.u., respectively.

2.2.4. PVDG power production constraint

The active power produced by the PVDG system should be bounded by the operating limits given in Eq. (17).

$$P_{PVDG}^{\min} \leq P_{PVDG} \leq P_{PVDG}^{\max} \quad (17)$$

where, $P_{PVDG}^{\min}$ and $P_{PVDG}^{\max}$ are lower and upper limits of the PVDG system output. These limits depend on the technical characteristics of the system and the weather conditions.

3. Loss sensitivity analysis for the optimal PVDG allocation

In this study, the appropriate location of the PVDG system is determined based on the sensitivity of lines losses to changes in the injections of the real power at the network nodes. This sensitivity can be measured by calculating the loss sensitivity factor (LSF) (Kc and Alkhwailidi, 2021). It has been shown in various studies that integrating LSF with other optimization methods enhances their computational efficiency and speed, especially for large scale RDSs and complex mixed-integer optimization problems (Kumar et al., 2022). A higher value of LSF at a given node signifies that a small change in power injection at that node will result in a significant change in the power losses within the network. Hence, nodes corresponding to higher LSF values are often considered strategic locations for DGs. Mathematically, the LSF can be expressed as follows (Werkie and Kefale, 2022).

$$LSF(i) = \frac{\partial P_{loss}(i-1, i)}{\partial P_i} = R(i-1, i) \left(2 \frac{P_i}{V_i^2} \right) \quad (18)$$

where i represents the node number.

It can be noted that by ordering the obtained LSF values in descending order, the node that remains at the top can be interpreted as the optimal location of the PVDG system. The steps of the LSF based method are detailed as follows.

- Step 1: Read line and bus data.
- Step 2: Run the system power flow and determine the active power injected at each bus using Eq. (1) and its voltage magnitude.
- Step 3: Calculate the LSF value of each bus using Eq. (18).
- Step 4: Arrange these obtained values in descending order.
- Step 5: Select the candidate bus that has the highest LSF value as the optimal PVDG system placement.

4. Proposed optimization method for the optimal PVDG sizing

After identifying the optimal location of the PVDG system using the LSF based method, the next step is to find the optimal capacity of the PVDG unit at the selected locations. This step involves the determination of the appropriate capacity of the PVDG unit that provides the maximum benefits, including minimum power losses and improved voltage profile. To address the challenges posed by nonlinear problems, researchers frequently adopt nature-inspired evolutionary algorithms. The BA is indeed one of the popular nature inspired optimization algorithms. Among these algorithms, the BA is prominent as a highly esteemed optimization approach. Unfortunately, like any other algorithm, BA has some common drawbacks. One of these drawbacks is that BA has various parameters such as loudness, pulse rate, and frequency that need to be carefully adjusted for optimal performance. This can be challenging and time-consuming. Moreover, theoretical foundations describing the operating principle of BA are limited and not well-established. To address these drawbacks, a self-adaptive Doppler compensation based BA ((SA-DC) BA) has been presented in (Shehab et al., 2023). Within the same context, an improved version of the (SA-DC)BA, called chaotic (SA-DC)BA, is developed in this section to further improve the algorithm performances.

4.1. Standard bat algorithm (BA)

The BA has been firstly discovered by Xin-She Yong in 2010 (Shehab et al., 2023). Owing to its flexibility and robustness in dealing with highly intricate problems, the BA has been broadly introduced in various engineering fields. It is noteworthy to highlight that BA is based on imitating the bats behavior's echolocation in finding their target. For further elucidation, micro bats naturally used echolocation to help them in seeking prey and in avoiding obstacles in complete darkness. By using frequency modulated signals ranged from 25 kHz to 150 kHz (Shehab et al., 2023) sent by micro bats as sound pulse, the target (prey) can be easily determined. Generally, bats can transmit around 10–20 sound pulses per second (Shehab et al., 2023). However, around 200 pulses every second can be sent when bats come near the target.

The BA principle begins by randomly generating the bat population by initializing each bat position as given by Eq. (19). Initial velocities of bats can be set at zero or small value (Meng et al., 2015). Note that the initial position vector of the i th bat is symbolized by

$$X_i^0 = [x_{i,1}^0 \quad x_{i,2}^0 \quad \cdots \quad x_{i,D}^0], \quad j \in \{1, 2, \dots, D\}$$

where

$$x_{i,j}^0 = X_j^{\min} + \gamma_{i,j} (X_j^{\max} - X_j^{\min}), \quad i \in \{1, 2, \dots, N_{pop}\} \quad (19)$$

where D and N_{pop} are the number of bats and search space dimension, respectively. $\gamma_{i,j} \in (0, 1)$ is a random number generated using a uniform distribution. $X_j^{\max}$ and $X_j^{\min}$ are the upper and lower limits of the j th decision variable.

Then, a new population will be generated by updating the positions, velocities V^{k+1} and pulse frequencies of the bats during iteration $(k+1)$, according to Eqs. (20)–(22).

$$f_i = f^{\min} + \mu(f^{\max} - f^{\min}) \quad (20)$$

$$V_i^{k+1} = V_i^k + f_i(X_i^k - X^*) \quad (21)$$

$$X_i^{k+1} = X_i^k + V_i^{k+1} \quad (22)$$

where X_i^k and V_i^k denote the position and velocity of the i th bat at iteration k . $f^{\min}$ and $f^{\max}$ are the lower and upper allowable bat frequencies, respectively. $\mu \in (0, 1)$ is a random number. X^* represents the current global best bat location.

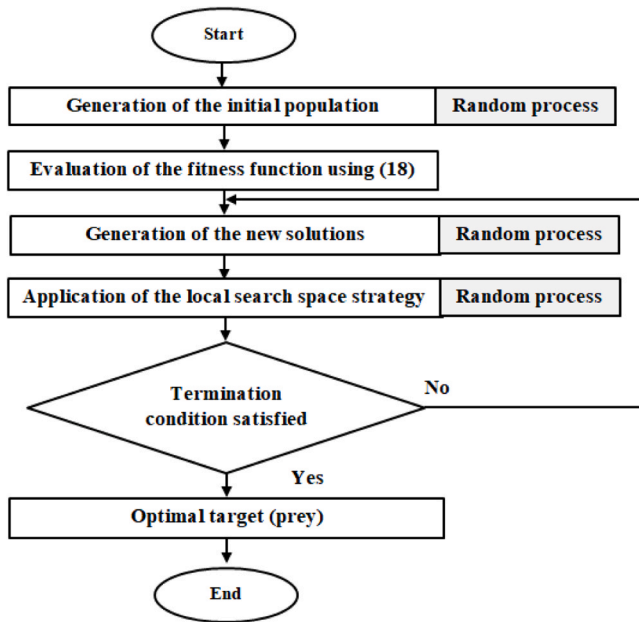


Fig. 2. Standard BA flowchart.

After generating the new population, a local search mechanism is applied. In this step, a new bat position is randomly generated around the best solution X^* according to a probability r_i , as illustrated in Eq. (23).

$$X_{new} = X^* + \varepsilon \times \hat{a}^k \quad (23)$$

where $\varepsilon \in (-1, 1)$ is a random number that guides local search direction. $\hat{a}^k$ represents the mean loudness of all bats at the k^{th} iteration.

After generating the new solution, pulse rate r_i and loudness a_i of the i^{th} bat are updated at each iteration, according to Eqs. (24) and (25). This step analyzes the bats foraging process.

$$a_i^{k+1} = \alpha \times a_i^k \quad (24)$$

$$r_i^{k+1} = r_i^0 [1 - e^{-\gamma^k}] \quad (25)$$

where $\alpha \in (0, 1)$ and $\gamma > 0$ are constants. $\alpha = \gamma$ is generally adopted in researches for the reason of simplicity (Huang et al., 2023). a_i^k and a_i^{k+1} are the loudness of the i^{th} bat at iterations k and $(k+1)$, respectively. $r_i \in (0, 1)$ is defined as the bat pulse rate. r_i^0 is the initial pulse rate of the i^{th} bat. Note that r_i^0 and a_i^0 can be generated randomly.

At the end of each iteration, the obtained bats are sorted according to the fitness function and the best solution X^* is extracted.

The flowchart of the standard BA is presented in Fig. 2.

4.2. Self-adaptive compensation for Doppler effect

To address the aforementioned BA limitations, several BA variants

Table 1
Chaotic maps description.

Number	Name	Formula	Parameters
1	Sinusoidal	$a_{k+1} = c \times a_k^2 \times \sin(\pi \times a_k)$	$c = 2.7$
2	Cubic	$b_{k+1} = \beta \times b_k \times (1 - b_k^2)$	$\beta = 2.59$
3	Chebyshev	$c_{k+1} = \cos \times (\text{acos}^{-1}(c_k))$	$\alpha = 8$
4	Quadratic	$d_{k+1} = d_k^2 - c$	$c = -2$
5	Tent	$e_{k+1} = \begin{cases} \rho e_k & \text{if } 0 \leq e_k \leq \frac{1}{2} \\ \rho(1 - e_k) & \text{if } \frac{1}{2} \leq e_k \leq 1 \end{cases}$	$\rho = 1$

have been introduced in the literature (Kouicem et al., 2022). For instance, in (Saxena et al., 2021), the bats' self-adaptive compensation mechanism for Doppler compensation is introduced in the original BA to further enhance its performance. This approach is adopted in this research study, owing to its ability to improve the standard BA in terms of biology compared to other BA variants which are looking for mathematically modifying the BA principle. It has been found and verified that this improved BA mostly outperforms the standard BA and other variants of the BA (Shehab et al., 2023). Indeed, in this approach, the standard BA is enhanced by incorporating the bats' habitat selection mechanism, introducing a novel search optimizer that leverages the Doppler effect in echolocation and investigating a self-adaptive local search space method. The bat's habitat selection involves selecting their quantum or mechanical behavior based on a randomly generated threshold P_x . Unlike the standard BA, where a constant frequency is adopted, the Doppler effect aims to modify and adjust the frequency of the bats' echolocation sound pulses as they move towards their prey.

4.2.1. Quantum behavior

The quantum evolutionary behavior is introduced to enhance the diversity of the algorithm population by enabling bats to forage across a vast range of habitats. Subsequently, the new bat position can be expressed as follows.

$$x_{ij}^{k+1} = \begin{cases} g_j^k + \emptyset \times |mean_j^k - x_{ij}^k| \times \ln\left(\frac{1}{u_{ij}^{k+1}}\right), & \text{if } rand < 0.5 \\ g_j^k - \emptyset \times |mean_j^k - x_{ij}^k| \times \ln\left(\frac{1}{u_{ij}^{k+1}}\right), & \text{if } rand \geq 0.5 \end{cases} \quad (26)$$

where g_j^k denotes the j^{th} dimension of the global best position at the k^{th} iteration. $\emptyset$ represents the self-adaptive linear decreasing coefficient (Saxena et al., 2021). This variable can be calculated as shown in Eq. (27). $mean_j^k$ is the mean of the j^{th} dimension of all bats' positions at the k^{th} iteration. It can be calculated as given by Eq. (28). u_{ij}^{k+1} is a uniform random value in the interval $(0, 1)$ not equal to zero.

$$\emptyset = \emptyset_{\max} - \frac{(\emptyset_{\max} - \emptyset_{\min})}{k_{\max}} k \quad (27)$$

where $\emptyset_{\max}$ and $\emptyset_{\min}$ are upper and lower values of $\emptyset$.

$$mean_j^k = \frac{1}{N_{pop}} \sum_{i=1}^{N_{pop}} x_{ij}^k \quad (28)$$

4.2.2. Mechanical behavior

By introducing the Doppler effect, the bat position x_{ij}^{k+1} and velocity v_{ij}^{k+1} as well as its sound rate frequency f_{ij} are slightly different from the aforementioned process in the basic BA presented in Eq. (19). In fact, by considering g_j^k as the target and by changing the i^{th} bat position x_i^k at each iteration k , near or far from the target, the self-adaptive compensation for Doppler effect can be efficiently explained in this part. If g_j^k is bigger than x_{ij}^k , bats will positively compensate for the echolocation's Doppler effect. Meanwhile, bats will negatively compensate for the Doppler effect in echolocations.

$$f_{ij} = f^{\min} + (f^{\max} - f^{\min}) \mu_{ij}^{k+1} \quad (29)$$

$$f_{ij} = \left(\frac{c + v_{ij}^k}{c + v_{ij}^k} \right) \left(1 + C_i^k \frac{(g_j^k - x_{ij}^k)}{|g_j^k - x_{ij}^k| + \varepsilon} \right) f_{ij} \quad (30)$$

$$v_{ij}^{k+1} = w v_{ij}^k + (g_j^k - x_{ij}^k) f_{ij} \quad (31)$$

Table 2

Allocation of chaotic maps to the random numbers.

Random variables	γ_{ij}	u_{ij}^k	μ_{ij}^k	C_i^k	α_{ij}^k
Adopted chaotic map	Sinusoidal	Cubic	Chebyshev	Quadratic	Tent
Length of the chaotic sequence	ND	KND	KND	KN	KND
Algorithm phases	Bat's initialization	Quantum behavior	Mechanical behavior	Mechanical behavior	Local search

$$x_{ij}^{k+1} = x_{ij}^k + v_{ij}^k \quad (32)$$

where C_i^k denotes the compensation rate of the i th bat at the k th iteration which is considered as a random value uniformly drawn from (0,1). c is the bat's speed in the air (340 m/s). v_{ij}^k is the velocity which corresponds to the global best position. μ_{ij}^{k+1} is a random number between 0 and 1. ε is a small value used to mitigate the zero-division-error. $w \in (0, 1)$ is the inertia weight.

4.2.3. Self-adaptive local search strategy

In this part, the new bat position, at each iteration k , is efficiently depending on the loudness a_i^k and it can be formulated using Eq. (33). In fact, bats are generally looking for decreasing the loudness to further convergence to the target and capture it. Accordingly, solutions are updated with a probability equal to r_x , as follows.

$$x_{ij}^{k+1} = g_j^k \times (1 + GD), \text{ if } \alpha_{ij}^{k+1} > r_x \quad (33)$$

$$GD = \text{randn}(0, \sigma^2) \quad (34)$$

$$\sigma^2 = |a_i^k - \hat{a}^k| + \varepsilon \quad (35)$$

where α_{ij}^{k+1} is a random value drawn from (0,1). GD represents the Gaussian distribution with the standard deviation σ^2 and mean 0. ε is a small value introduced to ensure that $\sigma^2 > 0$. $\hat{a}^k$ is the loudness average of all bats at iteration k .

4.2.4. Chaos theory

Evolutionary algorithm performance and convergence strongly depend on the random numbers involved in its various phases. In fact, introducing random sequences to initialize parameters can lead to different results at different processing times. In order to reach better results by avoiding local optimums, random variable should be well distributed (Afrabandpey et al., 2014). In this context, chaotic sequences can be used and applied to replace random number generators owing to their specific statistical and dynamical characteristics such as ergodicity, certainty and non-repetition (Huang et al., 2019). Hence, optimization algorithm using chaos theory have generally been improved and enhanced in terms of convergence performance and quality results compared to the original algorithms.

Chaotic systems can be mathematically formulated using chaotic sequences $\{x_k\}$ as follows.

$$x_{k+1} = f(x_k) \quad 0 < x_k < 1 \quad k = 1, 2, \dots, N_s \quad (36)$$

Where a_k represents the state variable and k is the number of iteration. N_s is the length of the chaotic sequence $\{x_k\}$.

Accordingly, there are various models to generate chaotic sequences. In this research study, five chaotic maps have been introduced and adopted to ameliorate the (SA-DC)BA performance and avoid local

optima convergence, namely the Sinusoidal (Si), Cubic (Cu), Chebyshev (Ch), Quadratic and (Qu) and Tent(Te) maps. Description and typical values of the parameters of these chaotic maps are presented in Table 1.

In the suggested optimization technique, random numbers γ_{ij} , u_{ij}^k , μ_{ij}^k , C_i^k , and α_{ij}^k involved in Eqs. (19), (26), (29), (30) and (33), respectively, are generated by the adopted chaotic map. Table 2 shows the allocation of the chaotic sequences to these random numbers as well as their lengths. It is to be noted that, the length of each chaotic sequence depends on the required number of random values to be generated which in turn depends on the bat's number (N), problem dimension (D) and maximum number of iteration (K).

As shown in Table 2, the Si, Cu and Te maps are incorporated in the bat's initialization, quantum behavior and local search phases of the proposed (SA-DC)BA, respectively. While, Ch and Qu maps are employed to generate the random numbers required for the mechanical behavior process.

After generating the different chaotic sequences, random numbers γ_{ij} , u_{ij}^k , μ_{ij}^k , C_i^k , and α_{ij}^k can be extracted from the chaotic sequences as given in Eqs. (37)–(41), respectively.

$$\gamma = \begin{bmatrix} a_1 & a_1 & \dots & a_D \\ a_{D+1} & a_{D+2} & \dots & a_{2D} \\ \vdots & \vdots & \vdots & \vdots \\ a_{(N-1)D+1} & a_{(N-1)D+2} & \dots & a_{DN} \end{bmatrix} \quad (37)$$

$$u^k = \begin{bmatrix} b_{\{(k-1)ND\}+1} & b_{(k-1)ND+2} & \dots & b_{\{(k-1)N+1\}D} \\ b_{\{(k-1)N+1\}D+1} & b_{\{(k-1)N+1\}D+2} & \dots & b_{\{(k-1)N+2\}D} \\ \vdots & \vdots & \vdots & \vdots \\ b_{\{kN-1\}D+1} & b_{\{kN-1\}D+2} & \dots & b_{kND} \end{bmatrix}, \quad k = 1, \dots, K \quad (38)$$

$$\mu^k = \begin{bmatrix} c_{\{(k-1)ND\}+1} & c_{(k-1)ND+2} & \dots & c_{\{(k-1)N+1\}D} \\ c_{\{(k-1)N+1\}D+1} & c_{\{(k-1)N+1\}D+2} & \dots & c_{\{(k-1)N+2\}D} \\ \vdots & \vdots & \vdots & \vdots \\ c_{\{kN-1\}D+1} & c_{\{kN-1\}D+2} & \dots & c_{kND} \end{bmatrix}, \quad k = 1, \dots, K \quad (39)$$

$$C^k = [d_{(k-1)N+1} \quad d_{(k-1)N+2} \quad \dots \quad d_{kN}], \quad k = 1, \dots, K \quad (40)$$

$$\alpha^k = \begin{bmatrix} e_{\{(k-1)ND\}+1} & e_{(k-1)ND+2} & \dots & e_{\{(k-1)N+1\}D} \\ e_{\{(k-1)N+1\}D+1} & e_{\{(k-1)N+1\}D+2} & \dots & e_{\{(k-1)N+2\}D} \\ \vdots & \vdots & \vdots & \vdots \\ e_{\{kN-1\}D+1} & e_{\{kN-1\}D+2} & \dots & e_{kND} \end{bmatrix}, \quad k = 1, \dots, K \quad (41)$$

where k is the actual iteration number. K , N and D are the maximum number of iterations, bat's number and problem dimension, respectively.

Note that,

$$\gamma_{ij} = \gamma(i, j) \quad (42)$$

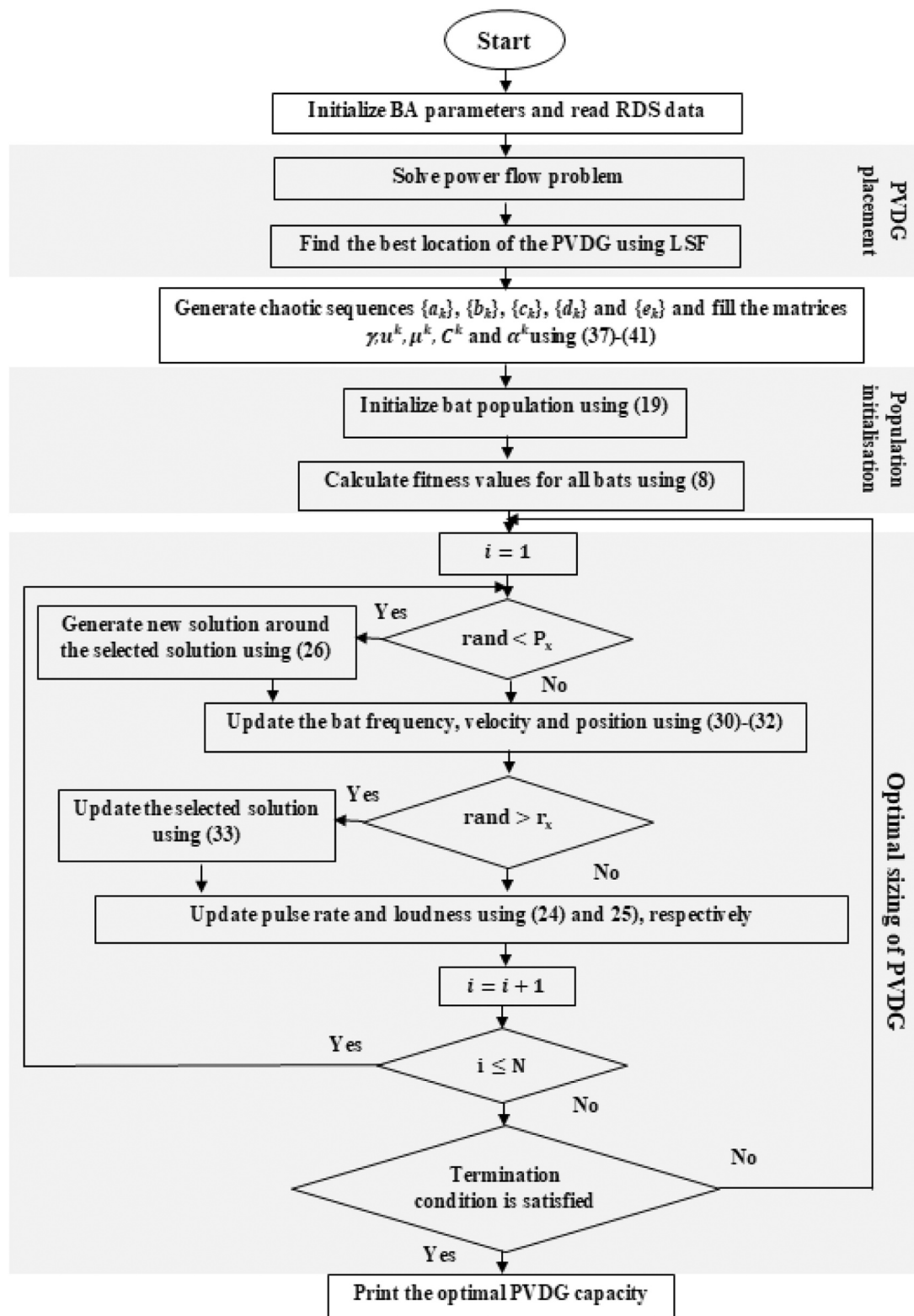


Fig. 3. (CSA-DC)BA flowchart.

Table 3
Unimodal benchmark functions.

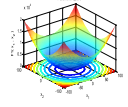
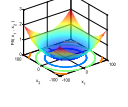
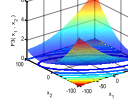
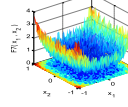
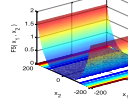
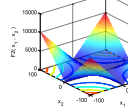
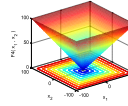
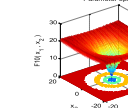
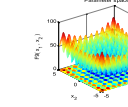
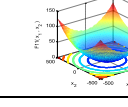
Function	Dim	Range	Optimum	Diagram
$F_1(x) = \sum_{i=1}^n x_i^2$	30	$[-100,100]$	0	
$F_2(x) = \sum_{i=1}^n x_i + 0.5 ^2$	30	$[-100,100]$	0	
$F_3(x) = \sum_{i=1}^n \left(\sum_{j=1}^i x_j \right)^2$	30	$[-100,100]$	0	
$F_4(x) = \sum_{i=1}^n i \times x_i^4 + rand(0, 1)$	30	$[-1.28,1.28]$	0	
$F_5(x) = \sum_{i=1}^{n-1} \left[100 \times (x_{i-1} - x_i^2)^2 - (x_i - 1)^2 \right]$	30	$[-30,30]$	0	
$F_6(x) = \sum_{i=1}^n x_i ^2 + \prod_{k=1}^n x_k $	30	$[-10,10]$	0	
$F_7(x) = \max_i \{ x_i \}, 1 \leq i \leq n$	30	$[-100,100]$	0	

Table 4
Multimodal benchmark functions.

Function	Dim	Range	Optimum	Diagram
$F_8(x) = -20 \times \exp \left(-0.2 \times \sqrt{\frac{1}{n} \times \sum_{i=1}^n x_i^2} \right) - \exp \left(\sqrt{\frac{1}{n} \times \sum_{i=1}^n \cos(2 \times \pi \times x_i)} \right) + 20 + e$	30	$[-32, 32]$	0	
$F_9(x) = \sum_{i=1}^n [x_i^2 - 10 \times \cos(2\pi \times x_i) + 10]$	30	$[-5.12, 5.12]$	0	
$F_{10}(x) = \left(\frac{1}{4000} \right) \sum_{i=1}^n x_i^2 - \prod_{k=1}^n \cos \left(\frac{x_i}{\sqrt{i}} \right) + 1$	30	$[-600, 600]$	0	

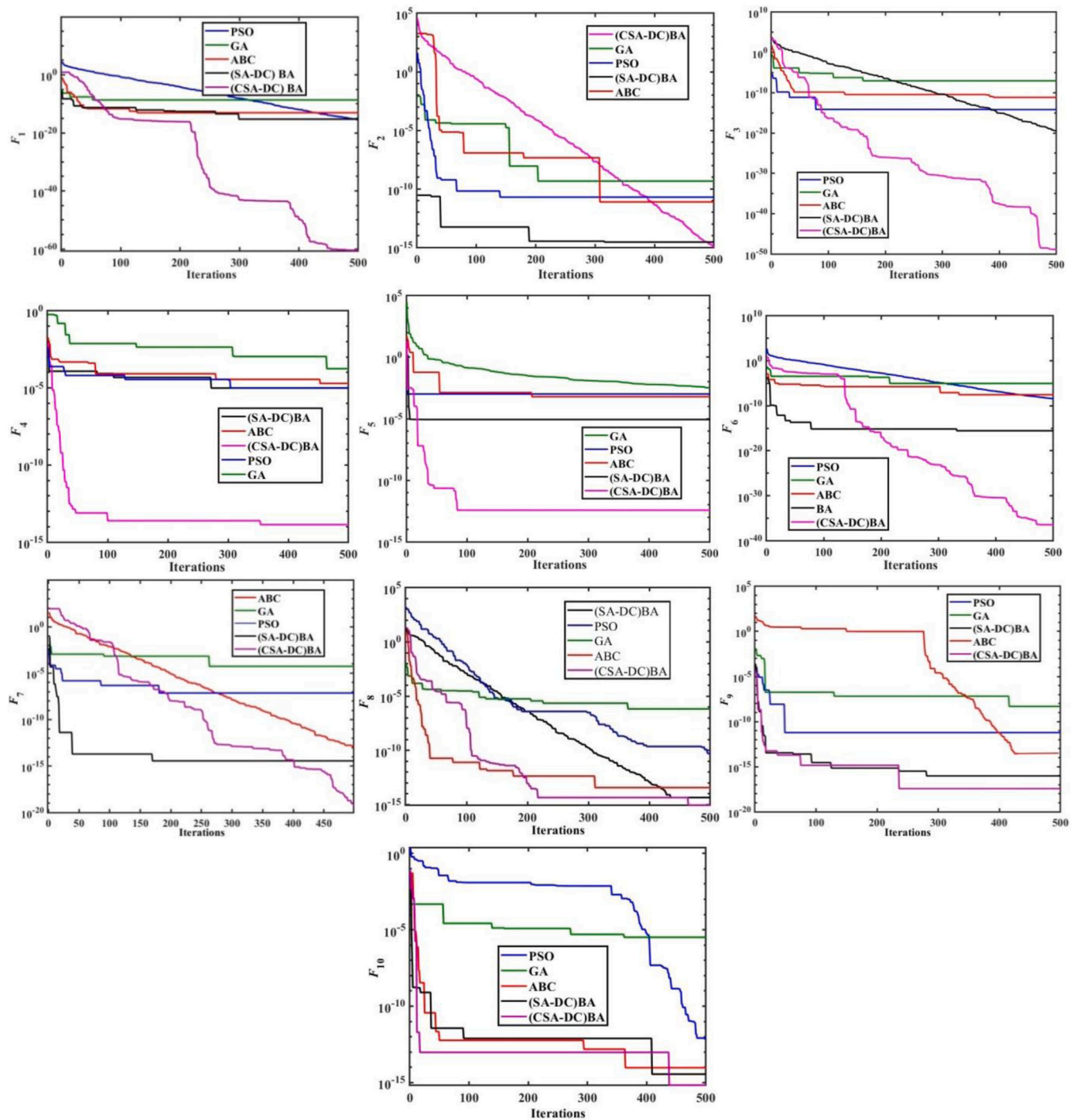


Fig. 4. Convergence characteristics of the benchmark functions.

$$u_{ij}^k = u^k(i, j) \quad (43)$$

$$\mu_{ij}^k = \mu^k(i, j) \quad (44)$$

$$C_i^k = C^k(i) \quad (45)$$

$$\alpha_{ij}^k = \alpha^k(i, j) \quad (46)$$

The flowchart and the pseudo-code of the proposed (CSA-DC)BA technique are presented in Fig. 3 and Algorithm 1, respectively.

Algorithm 1. Pseudo-code of the (CSA-DC)BA

detailed in Table 3. The second category is adopted to examine the proposed algorithm's exploitation capacity and to verify the local optima avoidance. These functions are defined as multimodal and presented in Table 4.

Fig. 4 shows the convergence characteristics of the proposed (CSA-DC)BA technique compared to PSO, GA, ABC and the improved (SA-DC)BA methods in minimizing the aforementioned benchmark functions.

According to Fig. 4, it can be observed that the comparative findings revealed that the (CSA-DC)BA yielded superior performance in terms of achieving appropriate outcomes convergence compared to the other comparative algorithms for both benchmark functions groups.

To further evaluate the robustness of the developed optimization

-
- (1) Initialization of BA parameters ($N, D, X_j^{min}, X_j^{max}, f^{min}, f^{max}, \hat{a}^k, P_x, r_x, K$).
 - (2) Generate the Si, Cu, Ch, Qu and Te chaotic sequences (Table 1).
 - (3) Construct matrices γ, u^k, μ^k, C^k and α^k using equations (37)–(41).
 - (4) Affect values for random variables $\gamma_{i,j}, u_{i,j}^k, \mu_{i,j}^k, C_i$, and $\alpha_{i,j}^k$ using equations (42)–(46).
 - (5) for $i=1$ to N do (Population initialization)
 - (5.1) for $j=1$ to D do
 - Generate bats position $x_{i,j}^0$ according to Equation(19).
 - (5.2) end for
 - (5.3.) Fitness evaluation using Equation (8).
 - (6) end for
 - (7) while $k < K$ do
 - (7.1) for $i=1$ to N do
 - (7.1.1) if $rand < P_x$ (Quantum behavior)
 - Generate the new solution using Equation (26).
 - (7.1.2) else (Mechanical behavior)
 - Update the bat frequency, velocity and position according to Equations (30), (31) and (32), respectively.
 - (7.1.3) end if
 - (7.1.4) if $rand > r_x$
 - Apply the self adaptive local search strategy according to (33).
 - (7.1.5) end if
 - (7.1.6) Calculate a_i^{k+1} and r_i^{k+1} using Equations (24) and (25), respectively.
 - (7.2) end for
 - (7.3) $k = k + 1$
 - (8) end while
 - (9) Determination of the optimal solution.
-

5. Numerical results using benchmark functions

In order to test and validate the effectiveness of the proposed (CSA-DC)BA compared to other optimization algorithms, ten benchmark functions, detailed in Tables 3 and 4, are selected and adopted. These functions can be classified into two main categories. The first group consists of functions with only one optimal solution, introduced to verify the meta-heuristic optimization technique's convergence performance and exploitation capacity. These functions are defined as unimodal and

method, statistical results including the mean and standard deviation (SD) values, are determined through thirty runs of the investigated optimization techniques. The obtained outcomes are tabulated in Table 5.

From Table 5, it can be seen that the proposed (CSA-DC)BA technique offers better results and reveals the most effective technique compared with the other comparative algorithms. After thoroughly testing the (CSA-DC)BA technique on various benchmark functions, it is implemented for the optimal sizing of PVDG units in RDSs.

Table 5
Experimental results.

	PSO		GA		ABC		(SA-DC)BA		(CSA-DC)BA	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
F1	3.12e−14	1.85e−14	3.19e−05	1.37e−05	8.04e−12	6.75e−12	3.76e−14	1.37e−14	9.07e−53	5.18e−53
F2	2.97e−08	1.54e−08	5.73e−08	3.06e−08	7.97e−12	3.41e−12	8.16e−14	4.61e−14	1.68e−14	7.18e−15
F3	5.00e−14	1.75e−14	1.66e−07	6.60e−08	3.60e−14	2.57e−14	2.23e−22	1.39e−21	4.25e−58	2.77e−58
F4	6.96e−06	3.70e−06	2.90e−02	1.10e−02	4.79e−06	1.20e−06	1.64e−06	8.86e−07	1.59e−14	3.96e−15
F5	3.80e−03	2.30e−04	8.63e−02	4.01e−02	7.50e−05	2.21e−06	2.42e−06	1.25e−07	1.97e−14	6.53e−15
F6	1.41e−09	1.04e−09	4.68e−04	3.20e−05	2.04e−05	7.82e−06	3.07e−14	1.92e−14	1.68e−20	1.59e−20
F7	8.56e−08	2.83e−08	6.58e−05	1.84e−05	9.30e−13	6.05e−14	5.15e−14	2.17e−14	3.72e−22	3.53e−22
F8	1.13e−12	4.95e−13	3.54e−08	1.26e−08	9.12e−14	1.58e−14	3.27e−14	1.12e−14	1.95e−15	5.15e−16
F9	8.25e−12	4.37e−12	7.24e−10	2.75e−10	1.05e−13	6.74e−15	2.90e−17	2.48e−18	2.89e−25	3.49e−26
F10	3.68e−11	2.31e−12	5.46e−06	2.38e−06	4.42e−13	4.29e−14	4.86e−13	4.68e−14	4.42e−18	5.70e−19

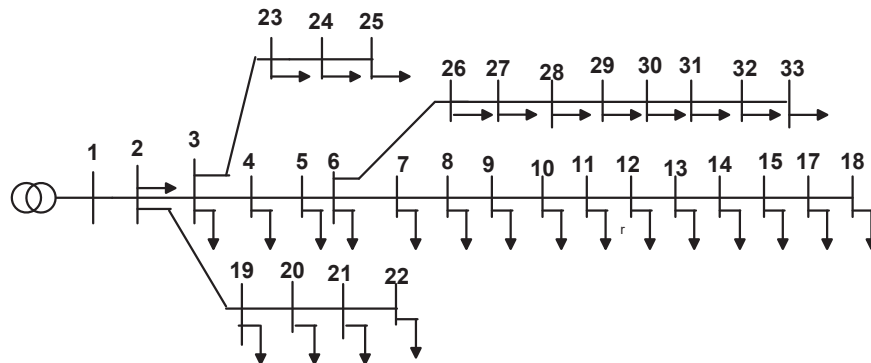


Fig. 5. Single line diagram of the IEEE 33 distribution network.

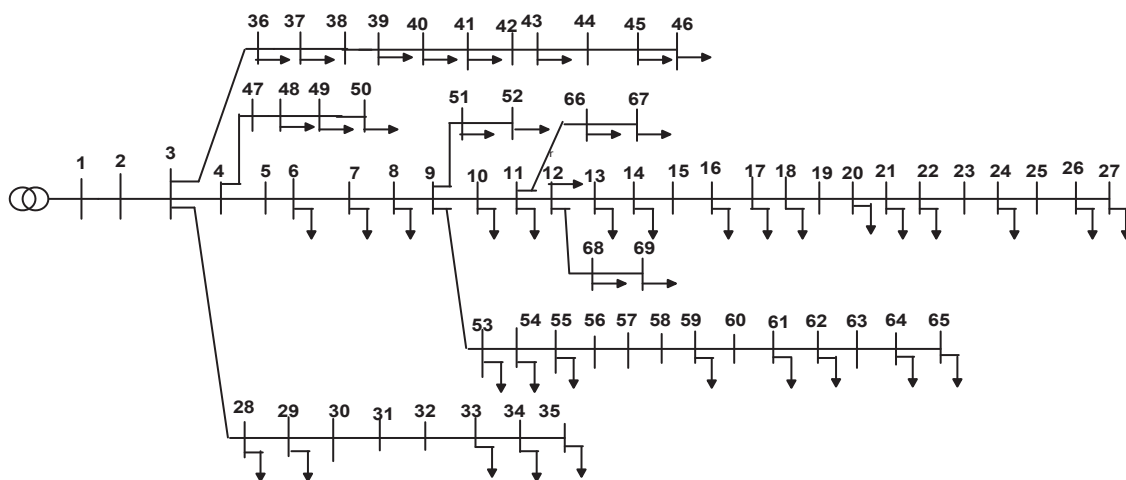


Fig. 6. Single line diagram of the IEEE 69 distribution network.

6. Simulation results

In this study, optimal sizing and placement of PVDG units in RDSs is described as a multi-objective optimization problem where both minimization of losses and improvement of voltage profile are considered. In order to apply the (CSA-DC)BA technique, problem objective functions are combined in one objective function using weighted-sum approach, as given in Eq. (8). The applicability and robustness of the suggested optimization method for solving this kind of problems are evaluated on two different test systems that are the standard IEEE 33-bus and 69-bus RDSs. The obtained results are compared with those of the same comparative algorithms used in the previous sub-section.

6.1. Studied systems

The IEEE 33-bus single line diagram is illustrated in Fig. 5, this radial distribution network is composed of 33 nodes numbered from 1 to 33 and 32 lines. Its base voltage is set at 12.66 kV. The total active and reactive powers consumed by loads are around 3.715 MW and 2.3 MVar, respectively (Abdelbadea et al., 2019). The initial total active power losses of this network is around 211 kW. However, the initial VD is around 1.6960 pu.

The IEEE69-bus single line diagram is shown in Fig. 6. This radial distribution network is composed of 69 bus numbered from 1 to 69 and 68 lines. The system voltage base and Volt-Ampere base are chosen to be 12 kV and 100 MVA, respectively. The total active and reactive system loads are 3.802 MW and 2.694 KVar, respectively (Latif and Palensky,

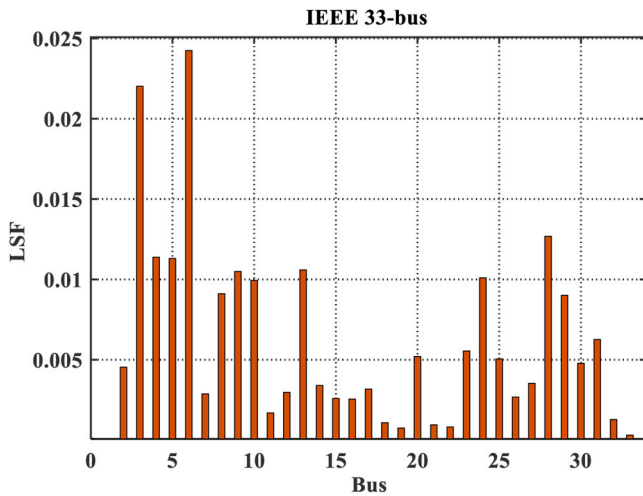


Fig. 7. LSF values of the IEEE 33-bus network.

2014). The initial total active power losses of this network is around of 225 kW. However, its initial voltage deviation VD is around of 1.8234 pu.

6.2. Optimal allocation of PVDG using the power loss sensitivity analysis

In order to find the optimal placement of the DG that leads to the minimum active power losses, a LSF based method is used. This analysis is based on finding the weak and sensitive node which leads to further reduce power losses. By arranging in descending order the obtained values of each bus LSF, node that presents the highest LSF value will be selected as the candidate bus for the optimal PVDG system location.

Figs. 7 and 8 show the values of LSF of all buses for the IEEE 33-bus and 69-bus systems, respectively. According to Fig. 7 the optimal placement of the PVDG system in the IEEE 33-Bus system is bus number 6. In fact, this node presents the highest LSF value. However, according to Fig. 8, the PVDG system will be integrated at bus 61 of the IEEE 69-Bus since it corresponds to the highest LSF value.

Once the best placement of PVDG unit is obtained for each system, the subsequent phase involves identifying its optimal size. To do this, the

suggested (CSA-DC)BA is applied for minimizing the objective function given by Eq. (8).

6.3. Optimal PVDG sizing using the proposed (CSA-DC)BA algorithm

The optimal PVDG capacity is obtained by minimizing total system losses and voltage deviation. This can be achieved by minimizing the weighted sum function described by Eq. (8). The set of the optimal Pareto solutions can be obtained by minimizing this function for different values of the weights ω_1 and ω_2 ranging from 0 to 1. This optimization problem is solved with the proposed (CSA-DC)BA algorithm and the aforementioned comparative algorithms. It is worth noting that minimum losses are obtained for $(\omega_1, \omega_2) = (1, 0)$. However, voltage deviation is minimized when $(\omega_1, \omega_2) = (0, 1)$. For fair comparison, the used optimization algorithms are run under same population size and maximum number of iterations. The values of these algorithms parameters are tabulated in Table 6. These values are adopted for both test systems.

6.3.1. Optimal sizing of PVDG in the IEEE 33-bus RDS

As previously mentioned, the optimal location of the PVDG unit in the IEEE 33-bus RDS is bus number 6. The optimal size of this unit is obtained using the (CSA-DC)BA method. To evaluate the performance of this method, obtained results are compared with those of the comparative algorithms. Convergence characteristics of these optimization methods are depicted in Fig. 9.

According to Fig. 9, it is clear that the (CSA-DC) BA converges much faster to the optimal solution compared to the other optimization techniques. Moreover, the minimum solutions in (a) as well as in (b) are reached using the proposed chaotic improved BA. This method converges to 102.7597 kW for $(\omega_1, \omega_2) = (1, 0)$ and to 0.2510 pu for $(\omega_1,$

Table 6

Optimization algorithms' parameters.

(SA-DC)BA	ABC	GA	PSO
$f^{\min} = 0;$	Acceleration	Mutation	Inertia weight =
$f^{\max} =$	coefficient = 1	probability =	0.9
0.5 $\phi_{\min} =$	Abandonment	0.2	Acceleration
0.5 $\phi_{\max} = 1.0$	limit = 1.2	Crossover	constants $C_1 =$
		probability =	$C_2 = 0.73$
		0.7	

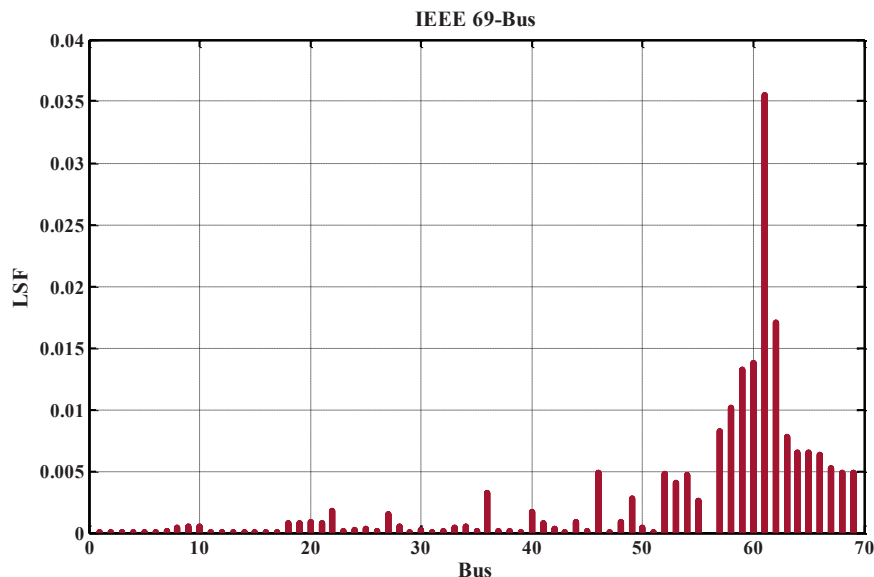


Fig. 8. LSF values of the IEEE 69-bus network.

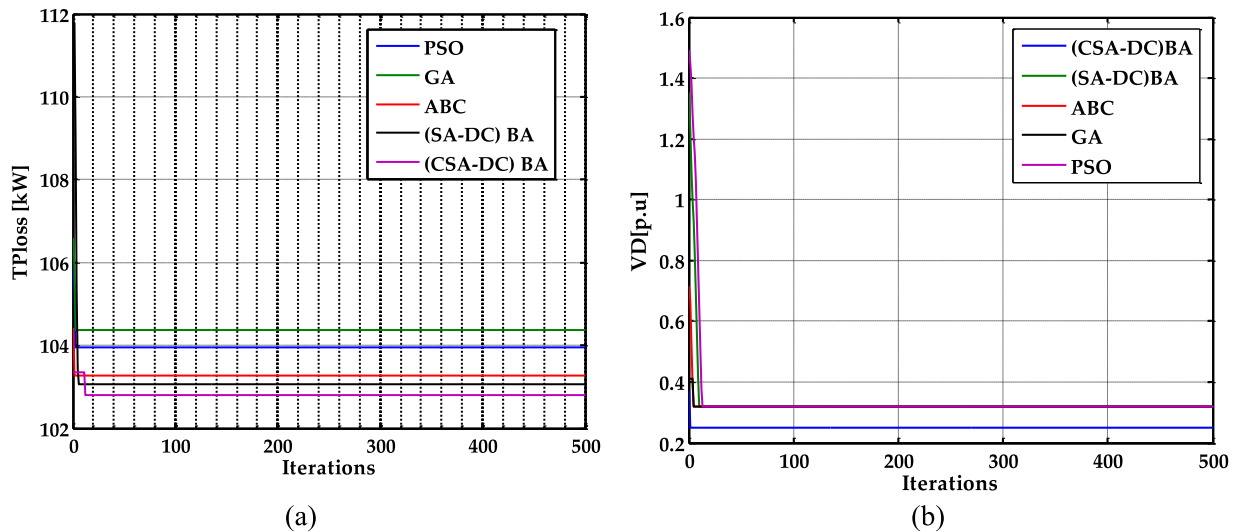


Fig. 9. Convergence characteristics (a) for $(\omega_1, \omega_2) = (1, 0)$; (b) for $(\omega_1, \omega_2) = (0, 1)$.

Table 7

Optimal non-dominated solutions obtained using (CSA-DC)BA method (IEEE 33-bus).

ω_1	ω_2	Ploss [kW]	VD [pu]	PVDG Size [MW]
1.00	0.00	102.7897	0.7924	2.59
0.95	0.05	102.7915	0.7890	2.6
0.90	0.10	102.8592	0.7685	2.66
0.85	0.15	103.1441	0.7379	2.75
0.80	0.20	103.3968	0.7209	2.8
0.75	0.25	103.7168	0.7039	2.85
0.70	0.30	104.1035	0.6869	2.90
0.65	0.35	104.5569	0.6699	2.95
0.60	0.40	104.6059	0.6682	2.96
0.35	0.45	105.0765	0.6530	3.0
0.50	0.50	107.8122	0.5854	3.20
0.45	0.55	111.5859	0.5180	3.40
0.40	0.60	113.8566	0.4844	3.50
0.35	0.65	119.1549	0.4174	3.70
0.30	0.70	125.4485	0.3507	3.90
0.25	0.75	127.1754	0.3340	3.95
0.20	0.80	127.8833	0.3274	3.97
0.15	0.85	128.6008	0.3207	3.99
0.10	0.90	128.9633	0.3174	4.00
0.05	0.95	136.7190	0.2510	4.20
0.00	1.00	136.7190	0.2510	4.20

$\omega_2) = (0, 1)$. By comparing the developed (CSA-DC)BA technique with other methods, it can be proved that this proposed method gives better outcomings in terms of power losses reduction and voltage deviation amelioration. As it can be seen, the proposed (CSA-DC)BA technique performed better than the four optimization techniques in terms of speed convergence and optimal solution determination.

Optimal non-dominated solutions obtained for various values of (ω_1, ω_2) using the proposed (CSA-DC)BA are tabulated in Table 7. From this table, it is obvious that minimum value of the total real power losses is obtained for $(\omega_1, \omega_2) = (1, 0)$, which is equal to 102.7597 kW. However, the minimum voltage deviation is obtained for $(\omega_1, \omega_2) = (0, 1)$, which is equal 0.2510 pu. This table also shows that the optimal values of the size of the PVDG unit for minimum losses and minimum voltage deviation are 2.09 pu. and 4.2 pu., respectively.

The trade-off curves provided by the investigated optimization algorithms are presented in Fig. 10. From this figure, it is apparent that the non-dominated solutions discovered by (CSA-DC)BA are well distributed across the entire trade-off surface compared to the other algorithms. Fig. 10 also shows that improving one objective might entail making concessions in the other objective. Hence, finding an optimal solution

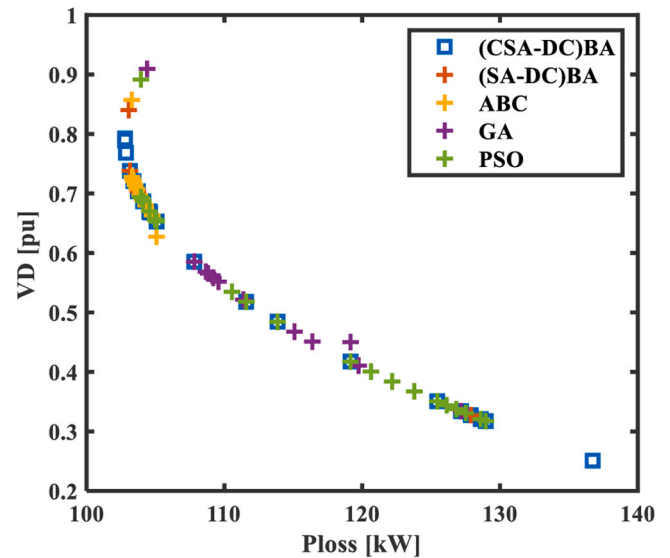


Fig. 10. Pareto fronts obtained by different optimization algorithms for IEEE 33-bus.

for the decision maker is a critical issue. In this study, the fuzzy-based method previously presented is implemented to find the best compromise solution. Table 8 presents the compromise solution data corresponding to the (CSA-DC)BA and the comparative algorithms. From this table, it can be seen that there is no compromise solution dominates the others. However, (CSA-DC)BA algorithm furnishes a compromise solution with the highest average satisfactory degree (ASD) among the other optimization methods which indicates that (CSA-DC)BA provides the most favorable compromise solution regarding the maker-decision satisfaction.

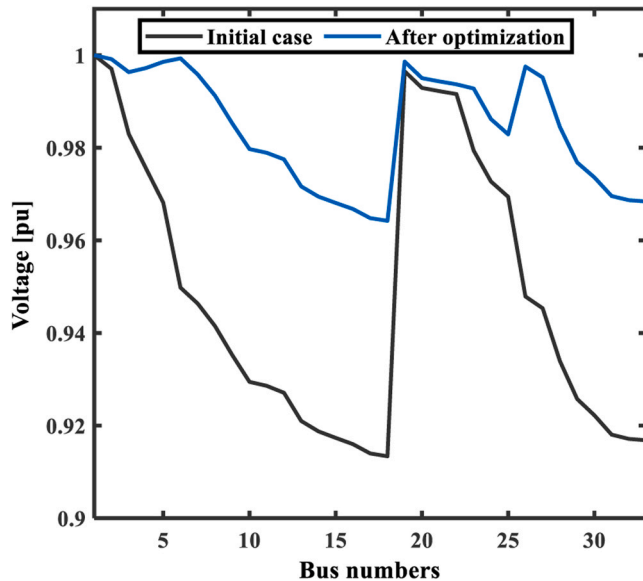
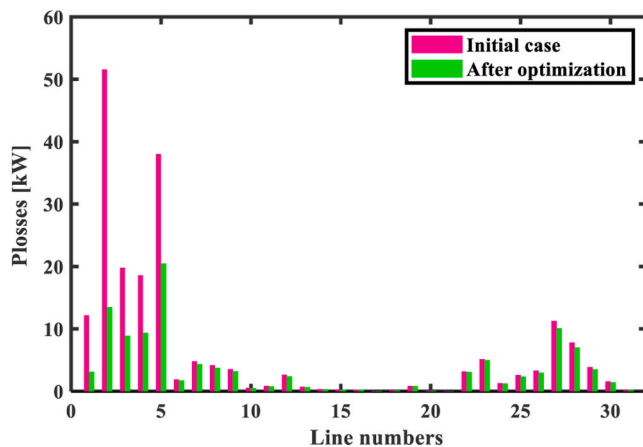
To assess the effectiveness of the proposed PVDG system in enhancing the voltage profile and reducing overall system losses, variations of nodes voltages and lines power losses before and after integration of the optimized PVDG are presented in Figs. 11 and 12, respectively. Note that the optimized PVDG corresponds to the compromise solution obtained using (CSA-DC)BA method.

According to Fig. 11, it is observed that the system voltage profile of the IEEE 33-bus RDS is improved after the integration of the PVDG system in the optimal position with the appropriate capacity compared

Table 8

Compromise solutions obtained by different optimization algorithms for IEEE 33-bus.

Algorithm	PVDG size (MW)	Ploss [kW]	DV [pu]	Losses reduction (%)	Voltage reduction (%)	ASD
(CSA-DC)BA	3.40	111.5859	0.5180	44.73	69.46	0.4159
(SA-DC)BA	3.00	105.0765	0.6530	47.95	61.5	0.4067
ABC	3.20	107.8122	0.5854	46.60	65.48	0.4024
PSO	3.00	105.0765	0.6530	47.95	61.50	0.4032
GA	3.20	107.8122	0.5854	46.6	65.48	0.4094
CSA-G (Latif and Palensky, 2014)	2.579	153,3	0.2	27.3	88	-

**Fig. 11.** Voltage profile before and after optimization.**Fig. 12.** Line power losses before and after integration of the PVDG system.

to that in the base case. In fact, as shown in the previous figure, the initial system voltage profile before integrating DG system is considered weak and poor because it is drastically located near the lower boundary of the acceptable limits given in Inequality (16), especially, for bus numbers 18, 31, 32 and 33. The initial voltage magnitudes of these nodes are 0.9134 pu, 0.918 pu, 0.9171 pu and 0.9168 pu, respectively. As a result, after the integration of the PVDG system the voltage magnitudes of these nodes are enhanced to 0.9642 pu, 0.9696 pu, 0.9687 pu and 0.9684 pu, respectively.

Consequently, it can be shown that the system voltage profile is significantly ameliorated and enhanced after the integration of the PVDG system. This proves that the proposed method offers better results

in improving the voltage profile.

Fig. 12 shows the lines power losses before and after the integration of the PVDG system. According to this figure, it can be observed that the power losses in the lines are significantly decreased after introducing the PVDG system. For instance, it can be observed that the power losses in the five first lines of the studied network are significantly decreased from 12.2 kW, 51.6 kW, 19.8 kW, 18.6 kW and 38 kW to 3.13 kW, 13.5 kW, 8.9 kW, 9.37 kW and 20.5 kW, respectively. That proves the efficiency of the proposed chaotic optimization technique in power loss reduction.

6.3.2. The IEEE 69-bus radial distribution network

The optimal location selected by the LSF analysis of the PVDG unit in the IEEE-69-bus RDS is at bus 61. The optimal size of this unit will be determined using the proposed (CSA-DC)BA method. To test the performance of this method, obtained results are compared with those of the comparison algorithms. Convergence characteristics of these optimization methods are depicted in Fig. 13.

According to Fig. 13, it is clear that the (CSA-DC)BA converges much faster to the optimal solution compared to the other optimization techniques. By comparing the developed (CSA-DC)BA technique with the other methods in terms of solutions quality, it can be seen that the proposed method gives better results in terms of power losses reduction and voltage deviation amelioration. For instance, the (SA-DC)BA method furnishes a minimum losses and minimum voltage deviation equal to 82.19 kW and to 0.5504 pu, respectively. However, (CSA-DC)BA achieves a minimum total losses of 81.5178 kW and a voltage deviation of 0.4874 pu, respectively. Indeed, (CSA-DC)BA harnesses chaos to introduce stochasticity, promoting a balance between exploration and exploitation, thereby contributing to a more robust and adaptable optimization process.

Optimal non-dominated solutions and the trade-off curves between the objectives achieved by the investigated optimization methods are tabulated in Table 8 and presented in Fig. 14, respectively. The non-dominated solutions are obtained by minimizing the objective function presented in Eq. (8) for various values of (ω_1, ω_2) . From Table 9, it is clear that an increase in the weight ω_1 corresponds to a greater emphasis on minimizing the voltage deviation function VD, while a decrease in ω_1 places more importance on minimizing the total real power losses. This dynamic relationship emphasizes the role of the weight values in fine-tuning the optimization process to prioritize one objective over another. To select a compromise solution among the Pareto solutions given in Table 9, the fuzzy-based method is also adopted in this case. Table 10 shows the compromise solutions data obtained by the (CSA-DC)BA and the other methods. By examining the ASD values of these compromise solutions, it can be clearly seen that the compromise solution extracted from the Pareto-front of (CSA-DC)BA method is more satisfactory since it corresponds to the highest value of ASD.

Fig. 15 represents the voltage profile of the IEEE 69-bus distribution network after the integration of the optimal PVDG system. It is to be noted that, the PVDG size corresponds to the compromise solution of the suggested (CSA-DC)BA method.

According to Fig. 15, it is observed that the system voltage profile is significantly improved after the integration of the optimized PVDG system compared to the case without PVDG system. Actually, the base

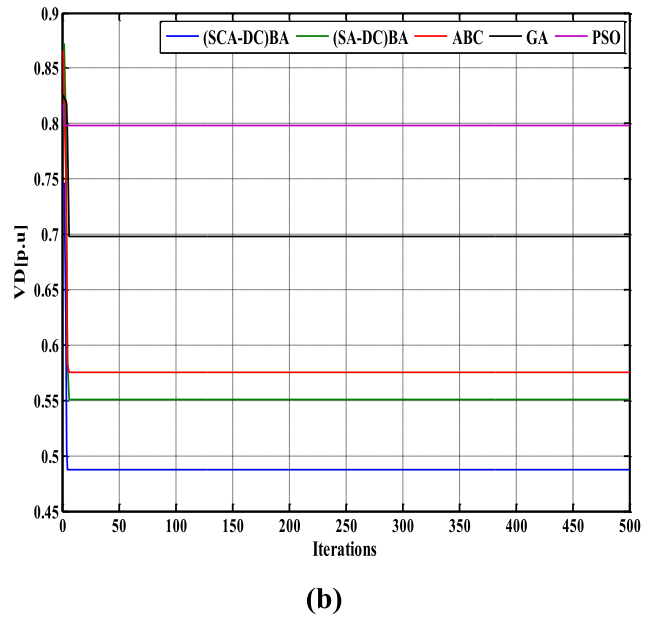
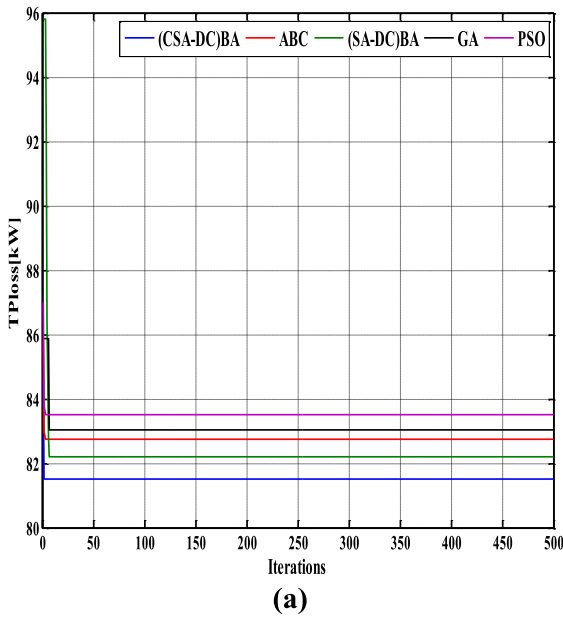


Fig. 13. Convergence characteristics. (a) for $(\omega_1, \omega_2) = (1, 0)$; (b) for $(\omega_1, \omega_2) = (0, 1)$.

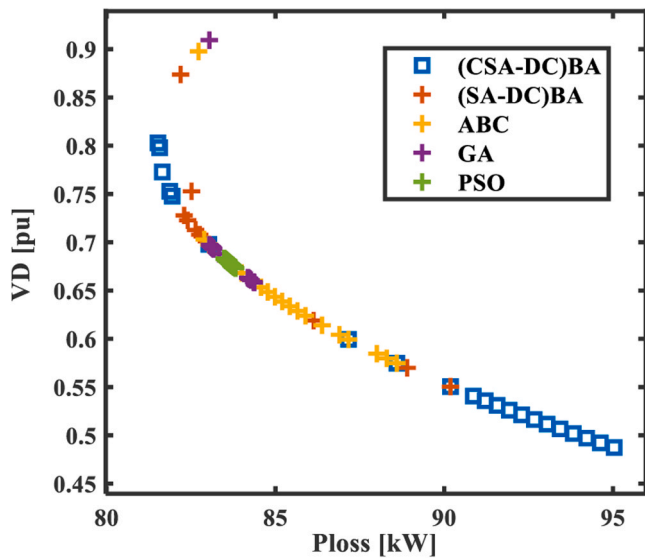


Fig. 14. Optimal Pareto fronts obtained by different optimization algorithms for IEEE 69-bus system.

case of the voltage profile is considered weak, especially, for bus number 65 located close to the lower boundary of the admissible limits given in Inequality (16). Indeed, it is equal to 0.9102 pu, which is considered near to 0.9 pu. The voltage magnitude at this node is ameliorated and enhanced after the integration of the optimized PVDG system to reach 0.9865 pu. Consequently, it can be concluded that the proposed chaotic method offers better results in terms of voltage profile improvement.

Fig. 16 shows the line power losses before and after the incorporation of the optimized PVDG system.

It is evident from Fig. 16 that power losses in the lines are reduced after introducing the PVDG system. Furthermore, it can be seen that this reduction is particularly noticeable in lines 5, 6 and 56 of the studied network. These line power losses are significantly reduced from 28.2 kW to 8.37 kW, from 29.3 kW to 8.69 kW and from 49.6 kW to 18 kW, respectively. That proves the efficiency of the proposed methodologies in terms of power losses reduction.

Table 9

Optimal non-dominated solutions obtained using (CSA-DC)BA method (IEEE 69-bus).

ω_1	ω_2	Ploss [kW]	VD [pu]	PVDG Size [MW]
1.00	0.00	81.5178	0.8029	1.89
0.95	0.05	81.5701	0.7979	1.9
0.90	0.10	81.6454	0.7728	1.99
0.85	0.15	81.8665	0.7528	2
0.80	0.20	81.9386	0.7478	2.1
0.75	0.25	83.0272	0.6979	2.2
0.70	0.30	87.1635	0.5993	2.3
0.65	0.35	88.5959	0.5748	2.35
0.60	0.40	90.1843	0.5504	2.4
0.55	0.45	90.8630	0.5407	2.42
0.50	0.50	91.2116	0.5358	2.43
0.45	0.55	91.5664	0.5310	2.44
0.40	0.60	91.9272	0.5261	2.45
0.35	0.65	92.2942	0.5213	2.46
0.30	0.70	92.6673	0.5164	2.47
0.25	0.75	93.0465	0.5116	2.48
0.20	0.80	93.4317	0.5067	2.49
0.15	0.85	93.8230	0.5019	2.5
0.10	0.90	94.2204	0.4971	2.51
0.05	0.95	94.6238	0.4922	2.52
0.00	1.00	95.0333	0.4874	2.53

6.4. Results Discussion

According to the simulation results, it can be concluded that the proposed (CSA-DC)BA is an efficient approach for optimal sizing and siting of PVDG units. This conclusion can be further proved through the comparative results presented in Tables 11 and 12. These tables show the optimal results obtained using the (CSA-DC)BA, (SA-DC)BA, ABC, GA, PSO, HPSO (Aman et al., 2014), MOPSO (Aman et al., 2013), DGPI (Memarzadeh and Keynia, 2020) and CBOM (Memarzadeh et al., 2023) algorithms for both IEEE 33-bus and IEEE 69-bus systems.

From these tables, it can be clearly seen that the (CSA-DC)BA algorithm outperforms the other optimization methods and presents better outcomes with significant minimization in power losses and important amelioration in the voltage profile. Indeed, it provides the lowest power losses and voltage deviation for both studied systems, in comparison with the other comparison algorithms. Moreover, after incorporation of the optimized PVDG, voltage magnitudes at all buses for both systems

Table 10
Compromise solutions for the IEEE 69-bus system.

Algorithm	PVDG size (MW)	Ploss [kW]	DV [pu]	Losses reduction (%)	Voltage reduction (%)	ASD
(CSA-DC)BA	2.3000	87.1635	0.5993	61 %	67.1 %	0.6144
(SA-DC)BA	2.0700	82.6309	0.7128	63.27 %	60.8 %	0.5249
ABC	2.0900	82.8885	0.7029	63.16	61.14 %	0.4808
PSO	2.1430	83.6982	0.6766	62.8 %	62.88	0.3385
GA	2.1010	83.0414	0.6974	63.09 %	61.7 %	0.4092

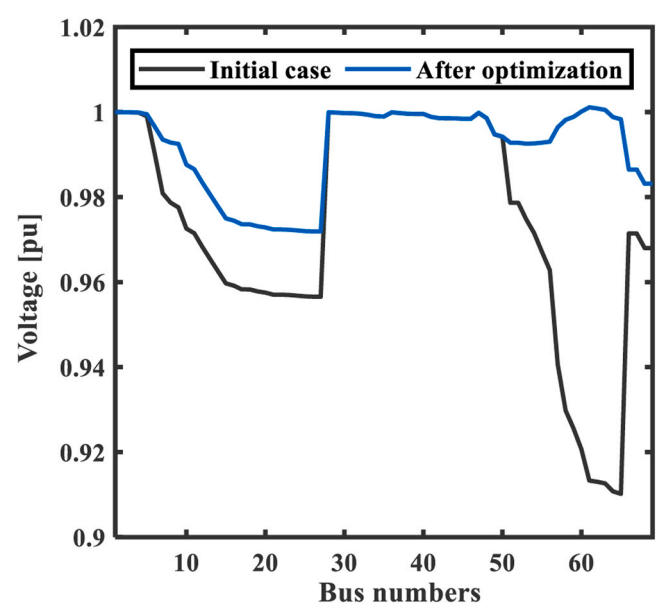


Fig. 15. Voltage profile of the IEEE 69-bus distribution network before and optimization.

are maintained within acceptable limits V_{min} and V_{max} .

7. Conclusion

In this paper, an efficient approach for optimal placement and sizing of PVDG units in RDSs is proposed. The suggested strategy begins by applying a power loss sensitivity analysis to find the optimal location of the PVDG units in the radial system. The most sensitive nodes are selected as strategic locations for PVDGs. After identifying the best

location of the PVDG unit, a new optimization algorithm incorporating the self-adaptive compensation for Doppler effect and chaos theory into the original BA for optimal sizing of the PVDG system is employed. The PVGD sizing problem is formulated as a two-objective optimization problem where total system losses and VD are the objective functions. In the proposed optimization algorithm, denoted by (CSA-DC)BA, the sine chaotic map is used to generate the random numbers required in the initialization step, the quantum behavior, the mechanical behavior and the local search process.

The performance of the proposed (CSA-DC)BA method is compared with other optimization algorithms in minimizing different unimodal and multimodal benchmark functions. Moreover, the applicability of the developed approach is assessed and tested on the standard IEEE 33-bus and 69-bus RDSs which are commonly utilized as benchmarks when addressing the optimal DG location and sizing problem. Obtained results showed that the optimized PVDG unit clearly contributed in the total power losses reduction and voltage profile improvement for both

Table 11
Comparative results for the IEEE 33-bus system.

Algorithm	Minimum losses [kW]	Minimum VD [pu]	V min [pu]	V max [pu]
(CSA-DC)BA	102.7897	0.2510	0.9642	1
(SA-DC)BA	103.1441	0.317	0.9507	1
ABC	103.3968	0.317	0.9589	1
GA	104.1035	0.317	0.9542	1
PSO	104.6059	0.317	0.9793	1
HPSO (Aman et al., 2014)	131.85	NA	0.979	1.05
MOPSO (Aman et al., 2013)	111.03	NA	0.9425	1
DGPI (Memarzadeh and Keynia, 2020)	142.2	NA	0.9482	1
CBOM (Memarzadeh et al., 2023)	103.97	NA	0.9511	1

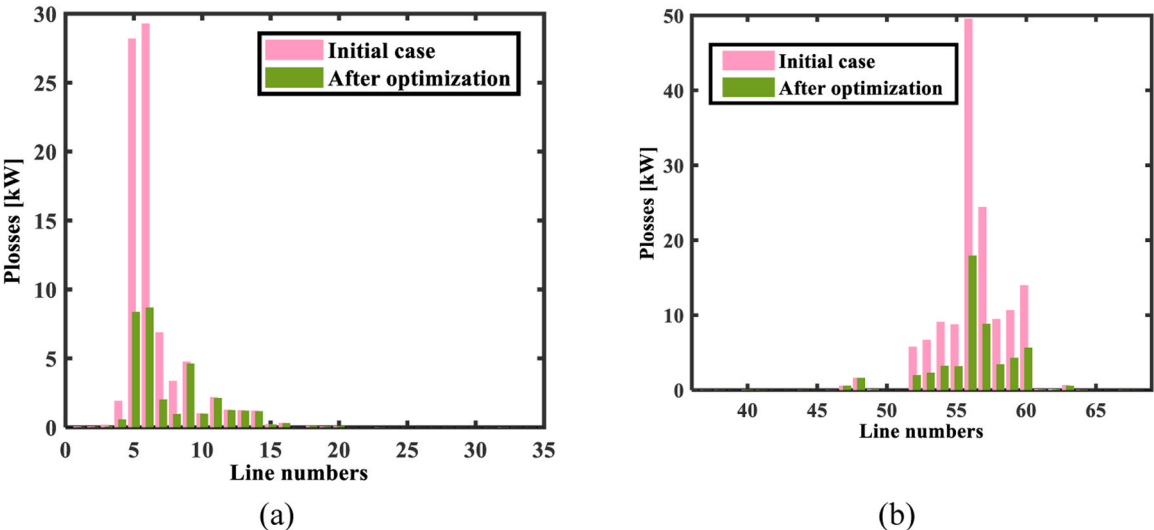


Fig. 16. Line power losses before and after integration of the PVDG system. (a) Lines 1–35; (b) Lines 36–68.

Table 12
Comparative results for the IEEE 69-bus system.

Algorithm	Minimum losses [kW]	Minimum VD [pu]	V min [pu]	V max [pu]
(CSA-DC)BA	81.5178	0.4874	0.9693	1
(SA-DC)BA	82.19	0.5504	0.9685	1
ABC	82.72	0.5748	0.9681	1
GA	83.04	0.6979	0.9680	1
PSO	83.52	0.7979	0.9678	1
HPSO (Aman et al., 2014)	87.13	NA	0.985	1.05
MOPSO (Aman et al., 2013)	84.04	NA	0.9692	1
DGPI (Memarzadeh and Keynia, 2020)	83.2	NA	0.9684	1
CBOM (Memarzadeh et al., 2023)	83.224	NA	0.9683	1

systems compared to the cases without PVDG unit.

Future work may focus on modeling the stochastic characteristic of the PV outputs by a chance constraint representing the probability that power balance constraints cannot be fulfilled. Then, chance-constrained programming can be used to find the optimal output of the PVDG system.

CRedit authorship contribution statement

Abdulaziz Almalag: Writing – original draft, Validation, Software, Methodology, Conceptualization. **Mansoor Alturki:** Writing – original draft, Methodology, Formal analysis, Conceptualization. **Rafik Neji:** Writing – original draft, Supervision, Methodology, Formal analysis, Conceptualization. **Khalid Alqunun:** Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **Tawfik Guesmi:** Writing – original draft, Validation, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Badr M. Alshammari:** Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization.

Author agreement statement

We the undersigned declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

No data was used for the research described in the article.

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